Aditya Reddy

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Education

Manipal University Jaipur — Jaipur, India

B.Tech, Electronics and Communication Engineering — *Aug 2024 – May 2028 (Expected)*

Fellowship

Samsung Fellowship – ISWDP, Cohort 8 — Jun – Jul 2026

IISc, Synopsys & Samsung Semiconductor India

- Selected for the Grade II Samsung Fellowship from 4,000+ applicants; completed ISWDP Levels 1 and 2 covering semiconductor device technology, TCAD workflows, scripted device creation, meshing, and advanced simulation.

Projects

INT8 Systolic-Array Accelerator | *SystemVerilog, LibreLane, OpenROAD, SKY130* — GitHub

- Took a 4×4 INT8 systolic-array accelerator from RTL to final layout on SKY130
- Met 200 MHz timing at all 9 PVT corners with zero DRC/LVS/antenna violations and 1.53 mV worst-case IR drop.
- Pipelined the datapath, doubling clock speed from 83 Mhz to 200 MHz for under 1% area (power: 12.7 to 69.6 mW).
- Built a post-route ECO flow closing timing at 82% utilization by buffering 45 critical endpoints.
- Extended LibreLane with custom Python steps for CTS tuning, ECO insertion, and per-corner reporting.

Dual-Clock Async FIFO + Formal Verification | *SystemVerilog, SymbiYosys, cocotb* — GitHub

- Designed a 64-deep, 16-bit asynchronous FIFO with Gray-coded pointers and CDC synchronization logic.
- Formally proved no overflow/underflow and full/empty exclusion via multiclock BMC
- Proved synchronizer invariants with k-induction and SVA corner-case assertions.
- Validated reset, wraparound, concurrent read/write, and asymmetric clock ratios with cocotb and Verilator.

RISC-V RV32IM Processor Core | *SystemVerilog, Vivado, Verilator* — GitHub

- Implemented a 5-stage RV32IM pipeline with EX/MEM forwarding, load-use hazard handling, and multiply/divide extension.
- Built hardware multiply/divide units, including a 32-cycle non-restoring radix-2 divider, with pipeline stalling through a ready handshake.
- Implemented dynamic branch prediction and performance counters for IPC, stalls, and branching behavior.
- Verified the core with SystemVerilog assertions across multiple test suites at a 100% pass rate.

Low-Latency Trading Gateway | *C++, FIX Protocol, UDP* — GitHub

- Built a C++ trading gateway that ingests UDP market data through a parser, order book, strategy, and order-execution pipeline.
- Minimized latency with lock-free queues, allocation-free memory pools, per-core CPU pinning, and a zero-copy state-machine FIX parser (1.4 μs average); achieved 10–15 μs average tick-to-trade latency (2.8 μs best case).

Activities

Volunteer | *Bison Asha School for Special Needs* — 2022 – 2024

- Supported classroom activities and organized inclusive events for students with special needs.

Technical Skills

Physical Design: RTL-to-GDSII, Floorplanning, Placement, CTS, Routing, STA, ECO, DRC/LVS, IR Drop, PVT

RTL & Verification: SystemVerilog, Verilog, SVA, Formal Verification, CDC, SymbiYosys, cocotb, Verilator

EDA Tools & PDKs: LibreLane, OpenROAD, Yosys, Magic, KLayout, OpenRAM, SKY130, Vivado

Programming: C, C++, Python, Tcl, Linux, Git